

Exercise 1

```
vihan@vihanzVM in ~ via C v16.1.1-gcc via v5.42.2 via v3.14.5
[*] / gcc -fopenmp -O2 -o ex1_barrier Ex1_barrier.c && OMP_NUM_THREADS=1 ./ex1_barrier
thread # 0
Outside Barrier # 0
Hi again from thread # 0
```

```
vihan@vihanzVM in ~ via C v16.1.1-gcc via v5.42.2 via v3.14.5
/ ) gcc -fopenmp -O2 -o ex1_barrier Ex1_barrier.c && OMP_NUM_THREADS=2 ./ex1_barrier
thread # 0
thread # 1
Outside Barrier # 0
Hi again from thread # 0
Outside Barrier # 1
Hi again from thread # 1
```

```
vihan@vihanzVM in ~ via C v16.1.1-gcc via v5.42.2 via v3.14.5
/ ) gcc -fopenmp -O2 -o ex1_barrier Ex1_barrier.c && OMP_NUM_THREADS=4 ./ex1_barrier
thread # 0
thread # 3
thread # 1
thread # 2
Outside Barrier # 3
Hi again from thread # 3
Outside Barrier # 0
Hi again from thread # 0
Outside Barrier # 1
Hi again from thread # 1
Outside Barrier # 2
Hi again from thread # 2
```

```
vihan@vihanVM in ~ via C v16.1.1-gcc via v5.42.2 via v3.14.5
) gcc -fopenmp -O2 -o ex1_barrier Ex1_barrier.c && OMP_NUM_THREADS=8 ./ex1
_barrier
thread # 1
thread # 2
thread # 3
thread # 4
thread # 7
thread # 5
thread # 0
thread # 6
Outside Barrier # 1
Outside Barrier # 3
Hi again from thread # 3
Outside Barrier # 4
Outside Barrier # 6
Outside Barrier # 2
Hi again from thread # 2
Hi again from thread # 6
Hi again from thread # 1
Outside Barrier # 5
Hi again from thread # 5
Outside Barrier # 7
Hi again from thread # 7

Outside Barrier # 7
Hi again from thread # 7
Outside Barrier # 0
Hi again from thread # 0
Hi again from thread # 4

vihan@vihanVM in ~ via C v16.1.1-gcc via v5.42.2 via v3.14.5
)
```

Q. Briefly describe how the pragma omp barrier command works in this program.

#pragma omp barrier makes every thread in the team wait at that point until all threads have arrived.

In this program, every thread prints its "thread #" line first but the order is unpredictable (thread scheduling isn't deterministic).

Once all threads hit the barrier, they're released together, and only then does any thread print the "Outside Barrier" / "Hi again" lines.

So you'll never see an "Outside Barrier" line before every thread has printed its first line - the barrier enforces that ordering as a hard synchronization point.